HSS ROBOTICS DAY · 18 OCTOBER

What will you build?

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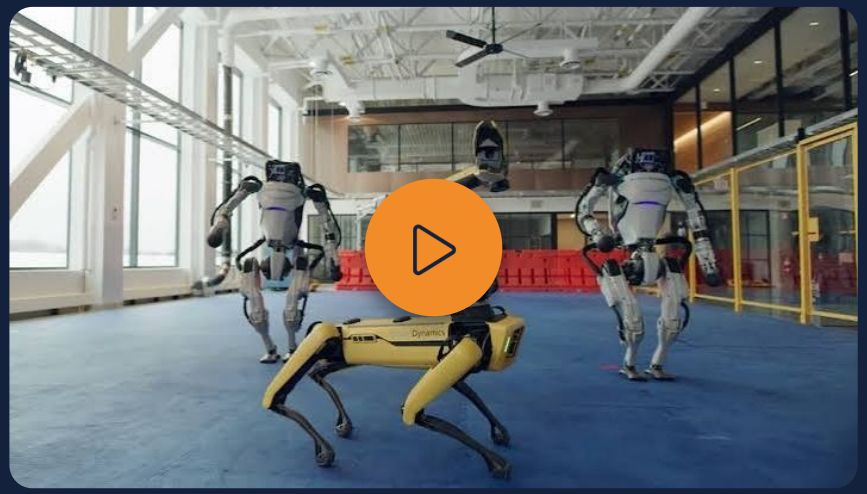


Who has a robot at home?

A robot vacuum. A smart speaker. An automatic door.

They're all robots.

Watch the video. What do these robots have to **notice** before they move?



Robots that dance

▶️ Play: Boston Dynamics, "Do You Love Me?"

So what is robotics?

Designing, building and coding machines that can do jobs by themselves.

Fun fact: "robot" comes from the Czech word *robota*, meaning hard work.



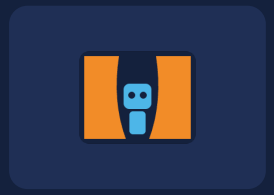
Engineering
Building the body

Electronics
Wiring the senses

Coding
Writing the brain

Maths
Distances and angles

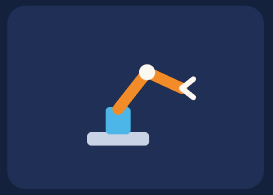
100 years of robots



1920

The word

A play invents the word "robot".



1961

Unimate

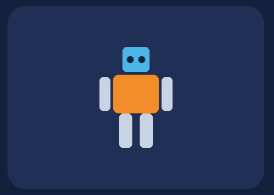
First factory robot arm, building cars.



1966

Shakey

First robot to plan its own moves.



1973

WABOT-1

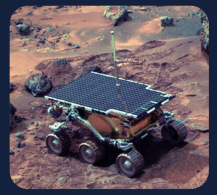
First full-size humanoid, in Japan.



1996

Honda P2

First humanoid to walk by itself.



1997

Sojourner

First rover to drive on Mars.



Today

Atlas

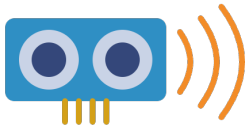
Robots that run, flip and learn.

HOW WE TELL ROBOTS WHAT TO DO

Record the moves → Computer code → Python (you!) → AI that helps write code

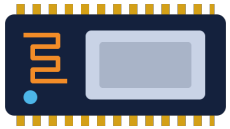
How are robots made?

Every robot does three things



Sense

Your **ultrasonic sensor** listens for echoes, like a bat.



Think

Your **ESP32** runs the MicroPython code you write.



Act

Your **servo motor** moves when the code says so.

In four weeks you'll build: 🚪 an automatic garage door · 📡 a radar

You're holding a moon computer… and more

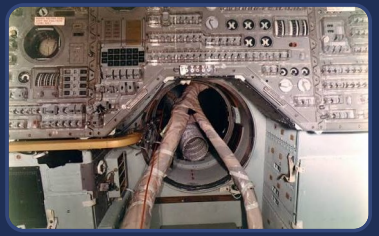
1969 · APOLLO GUIDANCE COMPUTER

Took astronauts to the Moon

~1
MHz speed

4
KB of memory

No Wi-Fi. Built by a huge NASA team.



TODAY · YOUR ESP32

Costs a few pounds

240
MHz speed

520
KB of memory

Wi-Fi and Bluetooth built in. Built by you.



Wi-Fi means your robot can talk to your phone, the internet and other robots.

Robots already do amazing jobs



A robot lands on Mars

▶️ Play: NASA Perseverance landing

Space

Rovers explore planets no person has visited.

Hospitals

Robot arms help surgeons make tiny, steady moves.



Farms

Robots pick fruit and pull weeds.

Rescue

Robots search places too dangerous for people.

BEIJING, CHINA · AUGUST 2026

The Robot Olympics are real

2,056	666	16	51
robots	teams	countries	events

Faster than Usain Bolt?

A robot called Tiangong Ultra ran 100 m in **9.39 seconds**. Bolt's world record is 9.58.

Not just sport

Football, kickboxing and dance, plus real jobs: firefighting, rescue and serving food.

Around the world

RoboCup, held in Korea in 2026, wants a robot team to beat the World Cup champions by 2050.

Robots should be good and safe

Robots help us

- Doing dangerous jobs
- Helping doctors and people with disabilities
- Exploring space and the deep ocean
- Creating brand-new jobs

✓ A safe robot...	✗ An unsafe robot...
Has a big red stop button	Can't be stopped
Stops when its sensors aren't sure	Trusts a broken sensor
Is tested again and again	Was never properly tested
Asks before recording you	Spies without asking
Only its owner controls it	Can be taken over online

Big question: if a self-driving car makes a mistake, whose fault is it?

Robot Olympics



Your next four weeks

TODAY



Meet your parts

Wake up your ESP32 with MicroPython.

25 OCT



Sense and move

Wire up the sensor and the servo motor.

1 NOV



Garage door

It opens all by itself when a car arrives.

8 NOV



Build a radar

Sweep the room and spot what's there.

No experience needed. You'll learn every step by doing it.

Which one sounds like you?

The Builder
Loves wiring things up and making them move.

The Coder
Loves telling a computer exactly what to do.

The Inventor
Always has ideas and loves to draw them.

The Tester
Loves finding what breaks, then fixing it.

Start with a problem. What would YOU fix?

A plant that waters itself · a medicine reminder for grandma · a bin that says when it's full

YOUR MISSION

Build it. Test it. Share it.



Made in the UK

- **Ameca**, a humanoid robot from Cornwall
- **Ocado**'s warehouse robots
- **Shadow Robot** hands, made in London

Join a team

- FIRST LEGO League
- VEX Robotics
- F1 in Schools

Keep learning

- Code Club
- NCSC CyberFirst (ages 11–17)
- Share your projects online, with a grown-up,
- HSS UK Shakti Parivaar exhibit

The MicroPython you'll use today is free, was started in the UK, and is built by people all over the world who share their work.

A bit about me

Jayesh Patel

25 years keeping computers safe, including more than 10 as a Chief Information Security Officer.

Contributor at Checkgate Solutions, which builds apps to solve real world challenges.

I'm still a strategist & problem solver, just like you're about to be.

Your questions?



11Plus Mastermind

An app that helps kids practise for the 11+ exam.



C²TEM

Spots a company's weak spots before hackers do.



Resilivault

Helps businesses keep going when things go wrong.



MyMythoscan

One RedTeaming platform for every attack surface.
- Web, Cloud, Code, Network and AI security.